

Data Discovery Scan Report

Overview

```
{
  "summary": "DPDP-aware data discovery, profiling, quality, and RAID report",
  "run_id": "37a37571-deae-459e-a95a-78e60009274c",
  "scores": {
    "quality_score": 100.0,
    "risk_penalty": 46,
    "final_score": 54.0,
    "score_drivers": [
      {
        "factor": "pii_masking",
        "severity": "high",
        "penalty": 15,
        "detail": "public.customers.customer_name is NAME with masking_status=NOT_MASKED"
      },
      {
        "factor": "pii_masking",
        "severity": "high",
        "penalty": 15,
        "detail": "public.customers.email is EMAIL with masking_status=NOT_MASKED"
      },
      {
        "factor": "sample_sufficiency",
        "severity": "medium",
        "penalty": 8,
        "detail": "Insufficient data for reliable profiling"
      },
      {
        "factor": "sample_sufficiency",
        "severity": "medium",
        "penalty": 8,
        "detail": "Insufficient data for reliable profiling"
      }
    ],
    "score_components": {
      "data_quality": 100.0,
      "metadata_penalty": 0,
      "pii_masking_penalty": 30,
      "sample_sufficiency_penalty": 16,
      "findings_penalty": 0
    }
  }
}
```

Ingestion

```
{
  "source_type": "postgresql",
  "database": "Scan_db",
  "tables_scanned": [
```

```

"public.customers",
"public.orders"
],
"num_tables": 2,
"owners": [
"postgres"
],
"rows_sampled": {
"public.customers": 2,
"public.orders": 2
}
}

```

Source Inventory

```

{
"source_type": "postgresql",
"database": "Scan_db",
"tables": [
{
"schema": "public",
"table": "customers",
"qualified_name": "public.customers",
"owner": "postgres",
"columns": [
{
"name": "customer_id",
"type": "INTEGER",
"nullable": false,
"default": null,
"primary_key": true
},
{
"name": "customer_name",
"type": "TEXT",
"nullable": false,
"default": null,
"primary_key": false
},
{
"name": "email",
"type": "TEXT",
"nullable": true,
"default": null,
"primary_key": false
},
{
"name": "created_date",
"type": "DATE",
"nullable": true,
"default": null,
"primary_key": false
}
],
"primary_key": [

```

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"customer_id"
],
"foreign_keys": [],
"indexes": [
{
"name": "customers_email_key",
"columns": [
"email"
],
"unique": true
}
],
},
{
"schema": "public",
"table": "orders",
"qualified_name": "public.orders",
"owner": "postgres",
"columns": [
{
"name": "order_id",
"type": "INTEGER",
"nullable": false,
"default": null,
"primary_key": true
},
{
"name": "customer_id",
"type": "INTEGER",
"nullable": true,
"default": null,
"primary_key": false
},
{
"name": "order_date",
"type": "DATE",
"nullable": true,
"default": null,
"primary_key": false
},
{
"name": "order_amount",
"type": "NUMERIC",
"nullable": true,
"default": null,
"primary_key": false
},
{
"name": "order_status",
"type": "TEXT",
"nullable": true,
"default": null,
"primary_key": false
}
],
"primary_key": [

```

```

"order_id"
],
"foreign_keys": [],
"indexes": []
}
]
}

```

Data Quality

```

{
  "summary": "Profiled 2 table(s). Data quality score is 100.0.",
  "quality_dimensions": {
    "completeness": "Measured with cell-level null counts and high-null column detection.",
    "uniqueness": "Measured with duplicate sampled-row counts and ratios.",
    "consistency": "Measured with reusable column-pattern checks.",
    "accuracy": "Flagged where domain-like rules identify invalid values.",
    "integrity": "Supported by source primary key and foreign key metadata."
  },
  "table_wise_issues": [
    {
      "table": "public.customers",
      "issue": "Insufficient data for reliable profiling",
      "severity": "medium"
    },
    {
      "table": "public.orders",
      "issue": "Insufficient data for reliable profiling",
      "severity": "medium"
    }
  ],
  "overall_score": 100.0,
  "recommendations": [
    "Increase profiling sample size for public.customers before relying on quality conclusions.",
    "Increase profiling sample size for public.orders before relying on quality conclusions."
  ]
}

```

Column Intelligence

```

{
  "summary": {
    "columns_scanned": 9,
    "pii_columns": 2,
    "unmasked_pii_columns": 2,
    "high_risk_columns": 2,
    "sensitive_columns": 2
  },
  "tables": {
    "public.customers": [
      {
        "column": "customer_id",
        "data_type": "INTEGER",
        "nullable": false,

```

```

"pii_detected": false,
"pii_type": null,
"pii_confidence": "none",
"detected_by": null,
"masking_status": "NOT_APPLICABLE",
"recommended_masking": "NONE",
"classification": "Confidential",
"tags": [
"Customer Data"
],
"risk": "MEDIUM",
"evidence": "No PII pattern or column hint matched"
},
{
"column": "customer_name",
"data_type": "TEXT",
"nullable": false,
"pii_detected": true,
"pii_type": "NAME",
"pii_confidence": "medium",
"detected_by": "name_hint_engine",
"masking_status": "NOT_MASKED",
"recommended_masking": "tokenization",
"classification": "Sensitive",
"tags": [
"PII",
"Customer Data"
],
"risk": "HIGH",
"evidence": "2/2 sampled values looked name-like; column hint=True"
},
{
"column": "email",
"data_type": "TEXT",
"nullable": true,
"pii_detected": true,
"pii_type": "EMAIL",
"pii_confidence": "high",
"detected_by": "pattern_engine",
"masking_status": "NOT_MASKED",
"recommended_masking": "partial_email_mask",
"classification": "Sensitive",
"tags": [
"PII",
"Customer Data"
],
"risk": "HIGH",
"evidence": "2/2 sampled values matched EMAIL; column hint=True"
},
{
"column": "created_date",
"data_type": "DATE",
"nullable": true,
"pii_detected": false,
"pii_type": null,
"pii_confidence": "none",

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"detected_by": null,
"masking_status": "NOT_APPLICABLE",
"recommended_masking": "NONE",
"classification": "Confidential",
"tags": [
  "Customer Data"
],
"risk": "MEDIUM",
"evidence": "No PII pattern or column hint matched"
},
],
"public.orders": [
  {
    "column": "order_id",
    "data_type": "INTEGER",
    "nullable": false,
    "pii_detected": false,
    "pii_type": null,
    "pii_confidence": "none",
    "detected_by": null,
    "masking_status": "NOT_APPLICABLE",
    "recommended_masking": "NONE",
    "classification": "Public",
    "tags": [
      "Public"
    ],
    "risk": "LOW",
    "evidence": "No PII pattern or column hint matched"
  },
  {
    "column": "customer_id",
    "data_type": "INTEGER",
    "nullable": true,
    "pii_detected": false,
    "pii_type": null,
    "pii_confidence": "none",
    "detected_by": null,
    "masking_status": "NOT_APPLICABLE",
    "recommended_masking": "NONE",
    "classification": "Confidential",
    "tags": [
      "Customer Data"
    ],
    "risk": "MEDIUM",
    "evidence": "No PII pattern or column hint matched"
  },
  {
    "column": "order_date",
    "data_type": "DATE",
    "nullable": true,
    "pii_detected": false,
    "pii_type": null,
    "pii_confidence": "none",
    "detected_by": null,
    "masking_status": "NOT_APPLICABLE",
    "recommended_masking": "NONE",

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"classification": "Public",
"tags": [
  "Public"
],
"risk": "LOW",
"evidence": "No PII pattern or column hint matched"
},
{
  "column": "order_amount",
  "data_type": "NUMERIC",
  "nullable": true,
  "pii_detected": false,
  "pii_type": null,
  "pii_confidence": "none",
  "detected_by": null,
  "masking_status": "NOT_APPLICABLE",
  "recommended_masking": "NONE",
  "classification": "Confidential",
  "tags": [
    "Financial"
  ],
  "risk": "MEDIUM",
  "evidence": "No PII pattern or column hint matched"
},
{
  "column": "order_status",
  "data_type": "TEXT",
  "nullable": true,
  "pii_detected": false,
  "pii_type": null,
  "pii_confidence": "none",
  "detected_by": null,
  "masking_status": "NOT_APPLICABLE",
  "recommended_masking": "NONE",
  "classification": "Public",
  "tags": [
    "Public"
  ],
  "risk": "LOW",
  "evidence": "No PII pattern or column hint matched"
}
]
}
}

```

DPDP Compliance

```

{
  "status": "ACTION_REQUIRED",
  "pii_columns": 2,
  "unmasked_pii_columns": 2,
  "high_risk_count": 2,
  "required_actions": [
    "Apply masking, tokenization, or access restriction to unmasked PII columns.",
    "Validate purpose limitation and access controls for sensitive columns."
  ]
}

```

```
"Maintain traceability for detected PII and remediation ownership."
]
}
```

Data Findings

```
{
  "coverage": "Weak",
  "gaps": [
    {
      "type": "profiling",
      "description": "Insufficient data for reliable profiling",
      "evidence": "sample_size=2, threshold=30",
      "owner": "postgres",
      "severity": "medium"
    },
    {
      "type": "quality",
      "description": "public.customers.customer_name contains unmasked NAME PII.",
      "evidence": "2/2 sampled values looked name-like; column hint=True",
      "owner": "postgres",
      "severity": "high"
    },
    {
      "type": "quality",
      "description": "public.customers.email contains unmasked EMAIL PII.",
      "evidence": "2/2 sampled values matched EMAIL; column hint=True",
      "owner": "postgres",
      "severity": "high"
    }
  ],
  "data_issues": [
    "public.customers: Insufficient data for reliable profiling",
    "public.orders: Insufficient data for reliable profiling",
    "public.customers.customer_name contains unmasked NAME PII.",
    "public.customers.email contains unmasked EMAIL PII."
  ],
  "impact": [
    "Open metadata, ownership, integrity, or quality findings should be resolved before broad downstream consumption."
  ],
  "recommendations": [
    {
      "action": "Increase profiling sample size for public.customers before relying on quality conclusions.",
      "type": "data_fix",
      "priority": "medium",
      "owner": "data owner"
    },
    {
      "action": "Increase profiling sample size for public.orders before relying on quality conclusions.",
      "type": "data_fix",
      "priority": "medium",
      "owner": "data owner"
    }
  ]
}
```

```

    "action": "Apply masking or tokenization to public.customers.customer_name.",
    "type": "data_fix",
    "priority": "high",
    "owner": "postgres"
  },
  {
    "action": "Apply masking or tokenization to public.customers.email.",
    "type": "data_fix",
    "priority": "high",
    "owner": "postgres"
  }
]
}

```

RAID

```

{
  "risks": [
    {
      "description": "public.customers.customer_name contains unmasked NAME PII.",
      "severity": "high",
      "evidence": "2/2 sampled values looked name-like; column hint=True",
      "owner": "postgres"
    },
    {
      "description": "public.customers.email contains unmasked EMAIL PII.",
      "severity": "high",
      "evidence": "2/2 sampled values matched EMAIL; column hint=True",
      "owner": "postgres"
    }
  ],
  "assumptions": [
    {
      "description": "No explicit relationships were found in source metadata.",
      "validation_needed": "Confirm whether joins are enforced outside the database."
    }
  ],
  "issues": [
    {
      "type": "data",
      "description": "Low sample size for public.customers.",
      "severity": "medium",
      "impact": "Profiling confidence is reduced and rare data quality defects may be missed.",
      "owner": "postgres"
    },
    {
      "type": "metadata",
      "description": "Primary key is not defined for public.customers.",
      "severity": "medium",
      "impact": "Uniqueness and downstream joins may be ambiguous.",
      "owner": "postgres"
    },
    {
      "type": "data",
      "description": "Low sample size for public.orders.",

```

```

"severity": "medium",
"impact": "Profiling confidence is reduced and rare data quality defects may be missed.",
"owner": "postgres"
},
{
  "type": "metadata",
  "description": "Primary key is not defined for public.orders.",
  "severity": "medium",
  "impact": "Uniqueness and downstream joins may be ambiguous.",
  "owner": "postgres"
},
{
  "type": "quality",
  "description": "Insufficient data for reliable profiling",
  "severity": "medium",
  "impact": "May reduce reliability of reporting, analytics, or downstream processing.",
  "owner": "postgres"
},
{
  "type": "quality",
  "description": "public.customers.customer_name contains unmasked NAME PII.",
  "severity": "high",
  "impact": "Potential DPDP non-compliance and personal data exposure.",
  "owner": "postgres"
},
{
  "type": "quality",
  "description": "public.customers.email contains unmasked EMAIL PII.",
  "severity": "high",
  "impact": "Potential DPDP non-compliance and personal data exposure.",
  "owner": "postgres"
}
],
"dependencies": [],
"recommendations": [
  {
    "action": "Mask, tokenize, or restrict access to public.customers.customer_name.",
    "priority": "high",
    "owner": "postgres"
  },
  {
    "action": "Mask, tokenize, or restrict access to public.customers.email.",
    "priority": "high",
    "owner": "postgres"
  },
  {
    "action": "Increase profiling sample size for public.customers before relying on quality conclusions.",
    "priority": "medium",
    "owner": "data owner"
  },
  {
    "action": "Increase profiling sample size for public.orders before relying on quality conclusions.",
    "priority": "medium",
    "owner": "data owner"
  }
]

```

```

"action": "Apply masking or tokenization to public.customers.customer_name.",
"priority": "high",
"owner": "postgres"
},
{
"action": "Apply masking or tokenization to public.customers.email.",
"priority": "high",
"owner": "postgres"
}
]
}

```

Traceability

```

[
{
"table": "public.customers",
"column": "customer_id",
"detected_by": "classification_engine",
"pii_type": null,
"classification": "Confidential",
"risk": "MEDIUM",
"scan_run_id": "37a37571-deae-459e-a95a-78e60009274c"
},
{
"table": "public.customers",
"column": "customer_name",
"detected_by": "name_hint_engine",
"pii_type": "NAME",
"classification": "Sensitive",
"risk": "HIGH",
"scan_run_id": "37a37571-deae-459e-a95a-78e60009274c"
},
{
"table": "public.customers",
"column": "email",
"detected_by": "pattern_engine",
"pii_type": "EMAIL",
"classification": "Sensitive",
"risk": "HIGH",
"scan_run_id": "37a37571-deae-459e-a95a-78e60009274c"
},
{
"table": "public.customers",
"column": "created_date",
"detected_by": "classification_engine",
"pii_type": null,
"classification": "Confidential",
"risk": "MEDIUM",
"scan_run_id": "37a37571-deae-459e-a95a-78e60009274c"
},
{
"table": "public.orders",
"column": "customer_id",
"detected_by": "classification_engine",

```

```

"pii_type": null,
"classification": "Confidential",
"risk": "MEDIUM",
"scan_run_id": "37a37571-deae-459e-a95a-78e60009274c"
},
{
"table": "public.orders",
"column": "order_amount",
"detected_by": "classification_engine",
"pii_type": null,
"classification": "Confidential",
"risk": "MEDIUM",
"scan_run_id": "37a37571-deae-459e-a95a-78e60009274c"
}
]

```

Recommendations

```

[
{
"action": "Increase profiling sample size for public.customers before relying on quality conclusions.",
"type": "data_fix",
"priority": "medium",
"owner": "data owner"
},
{
"action": "Increase profiling sample size for public.orders before relying on quality conclusions.",
"type": "data_fix",
"priority": "medium",
"owner": "data owner"
},
{
"action": "Apply masking or tokenization to public.customers.customer_name.",
"type": "data_fix",
"priority": "high",
"owner": "postgres"
},
{
"action": "Apply masking or tokenization to public.customers.email.",
"type": "data_fix",
"priority": "high",
"owner": "postgres"
},
{
"action": "Mask, tokenize, or restrict access to public.customers.customer_name.",
"priority": "high",
"owner": "postgres"
},
{
"action": "Mask, tokenize, or restrict access to public.customers.email.",
"priority": "high",
"owner": "postgres"
}
]

```